

Owen Shi

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EDUCATION

Harvard College

Cambridge, MA

Current Freshman | B.A in Computer Science & Mathematics

Stuyvesant High School

Manhattan, NY

Stuyvesant Advanced Regents Diploma

SAT: 1570 (800 Math) | GPA: 97.45

WORK & LEADERSHIP EXPERIENCE

Stuy Fusion, FTC 479

Manhattan, NY

Vice President

Sept. 2023 - July 2026

- Engineered an **AprilTag computer vision pipeline** for field localization, odometry, and **autonomous alignment**.
- Implemented **multithreaded robot controls**, boosting loop frequency to **10 kHz** for low-latency response times.
- Migrated codebases to a **command-based architecture**, improving modularity, testing, and subsystem reuse.
- Developed and deployed the **team's website** to strengthen digital outreach, recruiting, and sponsor engagement.
- Designed beginner education workshops and delegated technical projects across **~32 members**.

BCTCio

Remote

Intern Full-Stack Web Developer

Feb. 2022 - Sept. 2025

- Delivered full-stack features for freelanced web apps using modern frameworks, databases, APIs, and cloud tools.
- Maintained a reusable **UI npm package** with **30+ production-ready components** for consistent client development.
- Created and maintained Future Sphere Education's **student portal**, serving **4,000+ active student users**, **300+ classes**, and **12+ teachers**. Aided in frontend and backend development.

NYC Department of Health and Mental Hygiene

Queens, NY

Information Technology Specialist Intern

June 2024 - Sept. 2024

- Helped system-wide (**3k+ computers**) recovery after global server outage by CrowdStrike. Imaged **62 PCs**.
- Modernized facility workstations by deploying and configuring upgraded monitors, keyboards, and peripherals.
- Audited hardware inventory and decommissioned laptops for compliant e-waste and asset lifecycle management.

PASSION PROJECTS

SpringForge | Python, VPython, Computational Geometry, Mechanical Kinematics, Calculus

- Developed a Python/VPython simulator modeling rotational motion in configurable multi-spring systems.
- Implemented **nonlinear torque calculations** and **numerical integration** for real-time angular motion.
- Built a polygon editor that computes **center of mass** and **inertia** for **custom shapes** via shoelace corollaries.

Slipstream | Java, Processing, Client-Server Networking

- Engineered a Java/Processing multiplayer racer with real-time **client-server synchronization** and state replication.
- Built pixel-based collisions, scrolling camera logic, and terrain-aware vehicle physics.
- Developed client-server networking to **synchronize opponent movement** and game state across machines.
- Implemented drifting, nitro, lap timing, HUD, and audio sound effects.

Bik | C, MIPS Assembly, CMake, Systems-Level Programming, Compiler/Interpreter Design

- Built a C compiler translating a custom high-level language into **executable MIPS assembly** with code generation.
- Designed a lexer, **recursive-descent parser**, AST, and scoped variable environment for end-to-end compilation.
- Implemented compile-time optimizations, including **constant folding**, to improve generated-code efficiency.

WORKFORCE SKILLS

- Languages:** English (fluent), Spanish (intermediate), & Mandarin Chinese (conversant)
- Technical Skills:** C, C++, Java, Kotlin, Lua, Go, Javascript, Typescript, Python, CMake, CSS, HTML, Next.js, TailwindCSS, PostgreSQL, Prisma, MongoDB, Node.js, React.js, MaterialUI, etc.
- Tools:** Visual Studio Code, Android Studio, Neovim, AWS, Google Cloud, Supabase, etc.